

Practical problems with equations



1 2

2

a $x + y = 13$

b 8 cm

3 3500 people

4 $x = 18$

5 18

6 48 hours

7 $\frac{36}{5}$ hours

8

a 2

b 9

9 10 minutes

10 4

11

a 8 mph

b 16 mph

12

a

	Distance (miles)	Rate	Time
Upstream	15	$n - 3$	$\frac{15}{n - 3}$
Downstream	30	$n + 3$	$\frac{30}{n + 3}$

b $n = 9$

13 $n = 81$



14

a 45 mph

b 66 mph

15 23 mph

16

a $(1 - k)t$

b

i $d = 380kt$

ii $d = 350(1 - k)t$

c $k = \frac{35}{73}$

d 1184 miles

17

a $x = 7$

b $x = 3$

c $x = 2$

18

a $m + 71 = 403$

b $m = 332$

19

a $p = 164$

b \$436

20

a $x + 2$

b $x + 4$

c $2x + 4$

21 10, 12, 14

22 13, 15, 17, 19

23

a $\frac{1}{4}x = 3x - 22$

b $x = 8$

24

a $189 = 21x + 105$

b $x = 4$

25

a $24x + 3$

b $x = 13$

26

a $2x - 3$

b 9 inches



c 15 inches

27 14, 15, 16

28

a $x = \frac{74}{9}$

b $\frac{296}{9}$

29 $x = 6$

30

a $x = -1$

b 8°

31 $x = 12$

32 $x = 5$

33

a $x = y + 1.36$

b $8x + 9y = 119$

c $y = 6.36$

d $x = 7.72$

34

a $x = y - 4$

b $2(x - 4) + 2y = 12$

c $y = 5$

d $x = 1$

35

a $x = 7y$

b $x + 2 = 5(y + 2)$

c $y = 4$

d $x = 28$

36 11°C

37

a 4

b 60

38

a \$271.12

b Underestimate

c \$1.73

d 32 years

e 2030

39 $B = \$14\,736.84$



40 17 years old

41 36 000 kg

42 28 advertisements

43 7 nights

44 240 L

45 10 years old

46
a 8

b 81

47 4 hours

48 $2w^2 + 13w = 260$

Additional questions

49

a $x = 2$

b $m = -3$

c $p = 4$

d $k = -9$

50

a $m = 7$

b $k = 5$

c $t = 1$

d $d = -7$

e $f = 4$

f $r = -1$

g $w = -3$

h $p = -5$

51

a $x = 8$

b $x = 20$

c $x = 75$

d No solution

52

a $h = 4$

b $x = 4$

c $x = 7$

d $m = 2$

e $x = -3$

f $p = -2$

g $k = 12$

h $x = 20$

$$i \quad x = 16$$

$$j \quad x = 11$$



$$x = 4$$

$$l \quad y = -3$$

$$m \quad x = 70$$

53

$$a \quad x = -5$$

$$b \quad x = \frac{2}{5}$$

$$c \quad x = -\frac{22}{19}$$

$$d \quad x = \frac{11}{4}$$

$$e \quad k = 8$$

$$f \quad y = 8$$

$$g \quad x = 22$$

54

a One solution

b Infinitely many solutions

c Infinitely many solutions

$$55 \quad x = 3$$

$$56 \quad x = 4$$

$$57 \quad x = 16$$

$$58 \quad x = 13 \text{ inches}$$

$$59 \quad p = \$186$$

$$60 \quad k = 14$$

61

- a Let x be the exact number of people at the concert. Stormie's guess was 800 too many, so Stormie's guess was $x + 800$. Pierce guessed 300 too few, so Pierce's guess was $x - 300$. The average of their guesses is 4250, so that means

$\frac{\text{Stormie's guess} + \text{Pierce's guess}}{2} = 4250$. We can then substitute in the equations we have found for Stormie's and Pierce's guesses to get the equation $\frac{x + 800 + x - 300}{2} = 4250$.

- b 4000

62

$$a \quad 12 = 0.5 \cdot L$$

Where L is the number of laps.

$$b \quad L = 24$$

- c Assuming both of them used their own cart, they would need to pay \$2 as a deposit. So



63 Bassam should have multiplied by $\frac{1 \text{ mi}}{1.6 \text{ km}}$ to convert from km to miles.

64

- a True. If you substitute the solution into the unknown variable and calculate each side separately, you should end up with the same value on both sides.
- b False. For example, the equation $x + 1 = 3$ has the solution $x = 2$, and the equation $x - 1 = 1$ also has the solution $x = 2$.
- c False. If the variable terms do not cancel out, then the equation must mathematically have one solution.
- d True. For example, if a value of x satisfies $2x + 3 = 1$, then it will also satisfy $4x + 6 = 2$.

65

- a
 - i $3n + 3 = 4n + 6$
 - ii One solution
- b
 - i $2n + 6 = 2n + 5$
 - ii No solution
- c
 - i $4n + 2 = 4n + 2$
 - ii Infinitely many solutions

66 $y = 8$

67 $B = \$15\,652.17$

68 $y = 16$ years old

69 $x = 12$

70

- a $x = 17$
- b 17, 18, 19
- c No, because the equation used to solve for x in part (a) has only one unknown, x , to the power of one. Therefore there is only one solution.